

Lab4Task3

test(ci): deliberately introduce data contract violation #1

Open

NeiwKai wants to merge 1 commit into main from test/deliberate-contract-failure

Conversation 0

Commits 1

Checks 2

Files changed 1

NeiwKai commented 2 minutes ago

No description provided.

test(ci): deliberately introduce data contract violation

eba98df

This branch has not been deployed

No deployments

Some checks were not successful

1 failing, 1 skipped checks

1 failing check
CI / test (pull_request) Failing after 57s

1 skipped check
CI / build (pull_request) Skipped 1 minute ago

No conflicts with base branch
Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

Unsubscribe

You're receiving notifications because you authored the thread.

1 participant

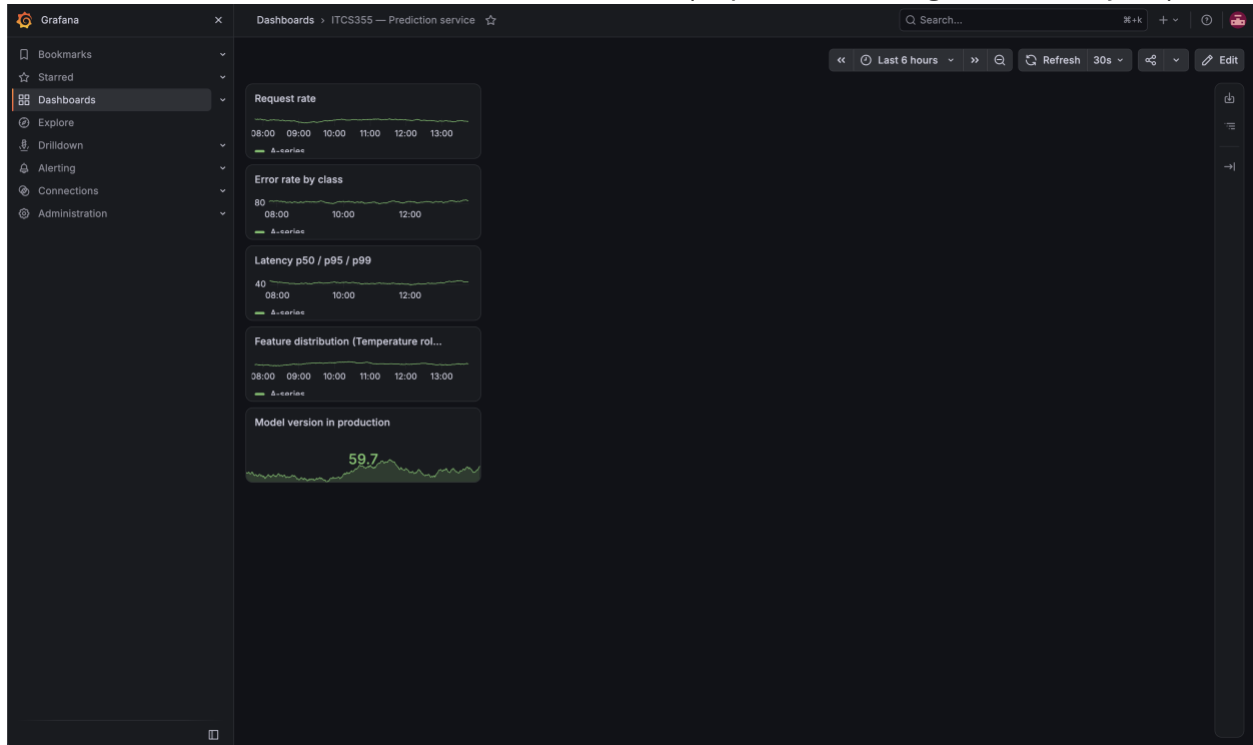
Data contract tests

1s

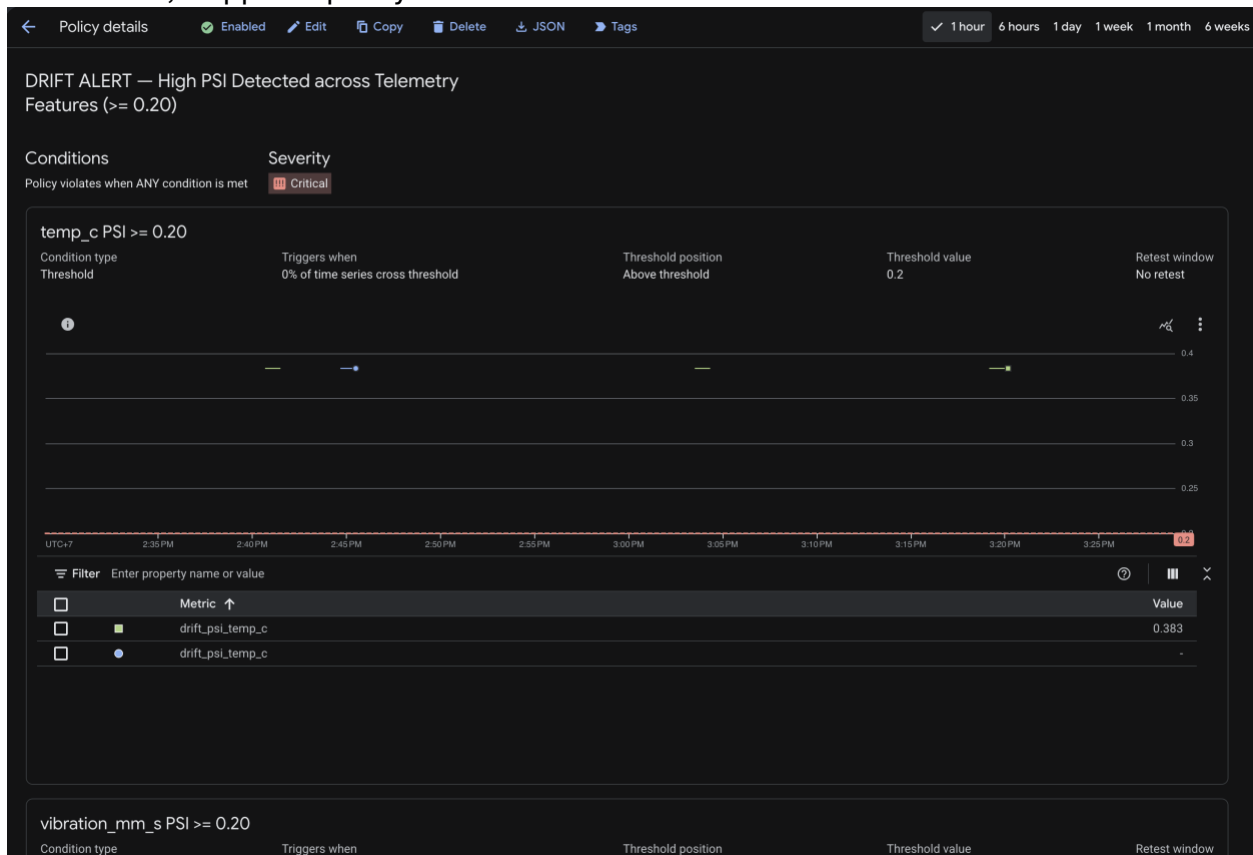
```
1 ▶ Run pytest -q tests/test_data.py [100%]
11 ..F.....
12 ===== FAILURES =====
13 test_features_within_plausible_ranges
14
15 df = reading_id machine_id ... ambient_humidity failed_within_7d
16 0 0 0 ... 63...65.351 1
17 5999 5999 239 ... 55.347 1
18
19 [6000 rows x 9 columns]
20
21 def test_features_within_plausible_ranges(df):
22     for col, (lo, hi) in data.PLAUSIBLE_RANGES.items():
23         assert df[col].min() >= lo, f"{col} below plausible floor: {df[col].min()}"
24     E AssertionError: temp_c below plausible floor: 48.186
25     assert np.float64(48.186) >= 150.0
26     E + where np.float64(48.186) = min()
27     E + where min = 0 82.478\n1 98.363\n2 94.514\n3 86.374\n4 89.962\n ... \n5995 106.808\n5996 110.740\n5997
28 95.909\n5998 104.880\n5999 110.776\nName: temp_c, Length: 6000, dtype: float64.min
29 tests/test_data.py:50: AssertionError
30 ===== short test summary info =====
31 FAILED tests/test_data.py::test_features_within_plausible_ranges - AssertionError: temp_c below plausible floor: 48.186
32 assert np.float64(48.186) >= 150.0
33 + where np.float64(48.186) = min()
34 + where min = 0 82.478\n1 98.363\n2 94.514\n3 86.374\n4 89.962\n ... \n5995 106.808\n5996 110.740\n5997 95.909\n5998
35 104.880\n5999 110.776\nName: temp_c, Length: 6000, dtype: float64.min
36 1 failed, 9 passed in 0.42s
37 Error: Process completed with exit code 1.
```

Suppanat Timngarm 6688132

Lab4 Task 4, dashboard. Prometheus + Grafana + (import monitoring/dashboard.json)



Lab4 Task 5, snippet of policy.



Suppanat Timngarm 6688132

Lab 4 Task 5, email alert fired.

INJECTION_TIME: 2026-10-09T08:19:11Z

REAL_EMAIL_ALERT: 2026-10-09T08:22:13Z

Take around ~3 minutes

10/9/26, 3:22 PM

Gmail - [ALERT - Critical] temp_c PSI >= 0.20 on itcs355-6688132



Suppanut Timngam

[ALERT - Critical] temp_c PSI >= 0.20 on itcs355-6688132

1 ข้อความ

Google Cloud Alerting <alerting-noreply@google.com>

9 ตุลาคม 2569 เวลา 15:22

ถึง:



[VIEW INCIDENT](#)

This is an automated notification set up by a Google Cloud user, not an official message from Google. If you weren't expecting this, please ignore it. [Learn more about Cloud Alerting.](#)

Alert firing Critical

Global - custom/itcs355/drift_psi_temp_c is above threshold of 0.2 with a value of 0.38333

Start time

Oct 9, 2026 at 8:22AM UTC (less than 1 sec ago)

Policy

[DRIFT ALERT — High PSI Detected across Telemetry Features \(>= 0.20\)](#)

Project

[itcs355-6688132](#)

Condition

temp_c PSI >= 0.20

metric : custom.googleapis.com/itcs355/drift_psi_temp_c course : itcs355

lab : 4 project_id : itcs355-6688132 student : itcs355-6688132

Policy documentation

Data drift detected on industrial telemetry features. One or more features crossed the 0.20 PSI threshold. Check Cloud Monitoring dashboard and inspect upstream sensor pipeline before retraining.

[VIEW INCIDENT](#)

Lab 4 Task 5,6; drift fired with timestamp

```

(ml_ops) mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_
mlaiops on mainxxx
# 1. Record injection start time
echo "INJECTION_TIME: $(date -u +"%Y-%m-%dT%H:%M:%SZ")"

# 2. Inject drift (+6.0 shift on temp_c)
make inject-drift

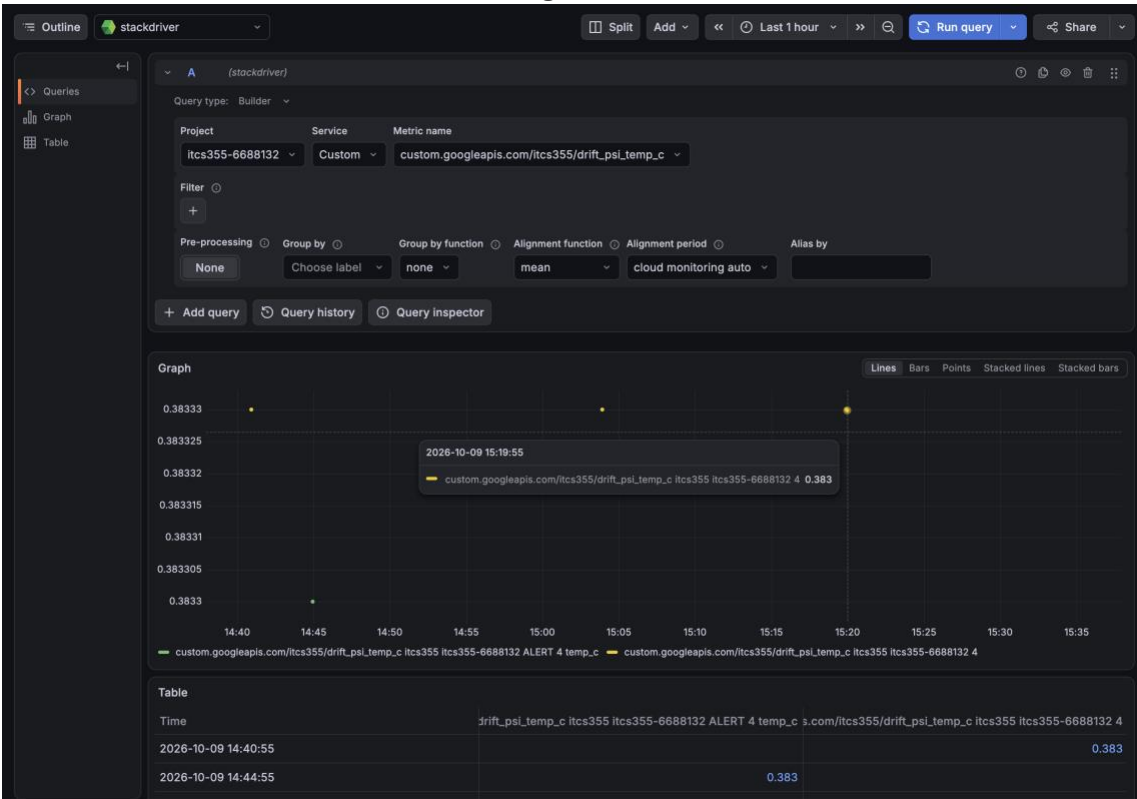
# 3. Record alert trigger time & run detector with metric emission
echo "ALERT_TRIGGER_TIME: $(date -u +"%Y-%m-%dT%H:%M:%SZ")"
make drift || true
INJECTION_TIME: 2026-10-09T08:19:11Z
python scripts/inject_drift.py --feature temp_c --mode shift --magnitude 6
wrote data/current.csv mode=shift feature=temp_c
mean 79.5800 → 85.5800

Now run: python -m monitoring.drift --current data/current.csv
Record the time from injection to alert. That is your detection latency.
ALERT_TRIGGER_TIME: 2026-10-09T08:19:12Z
python -m monitoring.drift --current data/current.csv --emit
feature                psi                ks verdict
temp_c                 0.38333    0.24567 significant
vibration_mm_s         0.00000    0.00000 stable
pressure_kpa           0.00000    0.00000 stable
hours_since_service    0.00000    0.00000 stable
load_pct               0.00000    0.00000 stable
ambient_humidity       0.00000    0.00000 stable
Cloud Monitoring: Emitted metric 'custom.googleapis.com/itcs355/drift_psi_temp_c'
= 0.3833
Cloud Monitoring: Emitted metric 'custom.googleapis.com/itcs355/drift_psi_vibrati
on_mm_s' = 0.0000
Cloud Monitoring: Emitted metric 'custom.googleapis.com/itcs355/drift_psi_pressur
e_kpa' = 0.0000
Cloud Monitoring: Emitted metric 'custom.googleapis.com/itcs355/drift_psi_hours_s
ince_service' = 0.0000
Cloud Monitoring: Emitted metric 'custom.googleapis.com/itcs355/drift_psi_load_pc
t' = 0.0000
Cloud Monitoring: Emitted metric 'custom.googleapis.com/itcs355/drift_psi_ambient
_humidity' = 0.0000

ALERT 1 feature(s) above threshold 0.2: temp_c
Before you retrain: is this drift, or is it a broken upstream pipeline? Retraini$

```

Lab4Task6, Grafana dashboard showing the drift



Lab4Task6, the log on google cloud console

